Rewaro Feedback Service API Documentation

1. Introduction

Purpose

The **Rewaro Feedback Service** is a serverless application designed to handle customer feedback efficiently, including **audio submissions** and **ratings**. By leveraging AWS-managed services, the system provides a scalable, cost-effective, and reliable architecture for **collecting**, **processing**, and **analyzing feedback data** through a RESTful API interface.

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2. Architecture Components

Core Services

- **API Gateway (rewaro-main-feedback)**:
  - Hosts a regional REST API for client interactions. This API is responsible for receiving feedback submissions through the `/process-feedback` and `/feedback-response-form` endpoints.
  - **Custom Domain**: `https://services-api-np.rewaro.in/feedback`
  - **Regional Endpoint**: `https://ibppac9wgg.execute-api.ap-south-1.amazonaws.com/PROD`
- **AWS Lambda Functions**:
  1. **feedback-service-audio-upload-to-SNS**:
     - Handles feedback submission (including audio or text review) from clients.
     - Validates the feedback data, stores it, and if audio is present, sends an SNS notification for processing.
  2. **feedback-service-audio-transcribe-upload-dynamo**:
     - Triggered by messages from the SQS queue.
     - Transcribes audio using **Amazon Transcribe** and stores the transcription along with the review data in DynamoDB.
  3. **feedback-service-feedback-response-form**:
     - Handles submissions of optional feedback forms for users who gave poor reviews (rating ≤ 2).
     - Collects name and contact information for follow-up.
- **Data Storage**:
  1. **Amazon S3**:
     - Securely stores uploaded audio files for processing.
  2. **Amazon DynamoDB**:
     - Maintains structured metadata and transcription results for real-time querying.
- **Message Orchestration**:
  1. **Amazon SNS**:
     - Coordinates asynchronous processing by notifying downstream components when a new feedback submission is received.
  2. **Amazon SQS**:
     - Ensures reliable delivery of transcription tasks to the Audio Transcriber Lambda.

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3. Architecture Flow

1. **Feedback Submission**:
   - The client sends a feedback submission (including optional audio) through the API Gateway.
2. **Feedback Processing**:
   - The **feedback-service-audio-upload-to-SNS** Lambda:
     - Validates the request payload.
     - Uploads the audio file to an Amazon S3 bucket.
     - Publishes a processing notification to an SNS topic.
3. **Message Queuing**:
   - The SNS topic forwards the notification to an SQS queue for transcription processing.
4. **Audio Transcription**:
   - The **feedback-service-audio-transcribe-upload-dynamo** Lambda:
     - Processes messages from the SQS queue.
     - Transcribes the audio file using Amazon Transcribe.
     - Stores transcription results and metadata in DynamoDB for querying.

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4. API Endpoints

POST /process-feedback

- **Description**: Submit customer feedback, including optional text or audio. The feedback should include a rating (1-5) and either a text review or an audio file (base64 encoded). If an audio file is provided, an SNS notification is triggered for transcription.
- **Request Parameters**:
  - **Query Parameter**:
    - `client_id`: The unique client identifier submitting the feedback (required).
- **Request Body**:
  1. **Only Rating and Empty Text Review**:

     ```
     {
       "rating": "1",
       "text_review": ""
     }
     ```

  2. **Rating and Text Review**:

     ```
     {
       "rating": "1",
       "text_review": "Bad Testtttt"
     }
     ```

  3. **Rating and Audio**:

     ```
     {
       "rating": "5",
       "audio": "data:audio/mp3; base64,T2dnUwACAAAAAAA"
     }
     ```

- **Responses**:
  1. **For Only Rating and Empty Text Review**:
     - **200 OK**:

       ```
       {
         "statusCode": 200,
         "message": "Rating recorded successfully.",
         "rating_id": "7153-1FU4-XXRQ-4CU2",
         "feedback_response": {
           "name": "Manish Kumar Tiwari",
           "phone_no": "+919910391614"
         },
         "isSocial": false
       }
       ```

  2. **For Rating and Text Review**:
     - **200 OK**:

       ```
       {
         "statusCode": 200,
         "message": "Rating and text review recorded successfully.",
         "rating_id": "809-AW74-LI0U-4ZGX",
         "feedback_response": {
           "name": "Manish Kumar Tiwari",
           "phone_no": "+919910391614"
         },
         "isSocial": false
       }
       ```

  3. **For Rating and Audio**:
     - **200 OK**:

```
{
  "statusCode": 200,
  "message": "Audio file uploaded and SNS notified.",
  "rating_id": "5478-6IX7-DWF0-4B0A",
  "feedback_response": {
    "name": "Manish Kumar Tiwari",
    "phone_no": "+919910391614"
  },
  "isSocial": false
}
```

POST /feedback-response-form

- **Description**: Submit an optional feedback response form for users who gave poor reviews (rating ≤ 2). Collects name and contact information for follow-up.
- **Request Body**:

```
{
  "name": "John Doe",
  "contact": "johndoe@example.com"
}
```

  - **name**: (string) The name of the user submitting the feedback.
  - **contact**: (string) Contact information for the user.

- **Responses**:
  - **200 OK**:

```
{
  "message": "Thank you for providing additional feedback."
}
```

  - **400 Bad Request**: Invalid input or missing fields.

```
{
  "error": "Invalid contact information."
}
```

  - **500 Internal Server Error**: Unable to process the form.

```
{
  "error": "Unable to process the form."
}
```

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5. AWS Resources

- **SNS Topic**:
  - Name: `user-feedback-topic`
  - Purpose: Sends notifications to SQS when new feedback is submitted that includes audio.
- **SQS Queue**:
  - Name: `UserFeedbackQueue`
  - Purpose: Receives SNS notifications for feedback processing and transcription tasks.
- **DynamoDB Table**:
  - Name: `CustomerFeedbackTable`
  - Purpose: Stores feedback metadata and transcription results.
  - Primary Key:
    - Partition Key: `client_id`
    - Sort Key: `rating_id`
- **S3 Bucket**:
  - Name: `customer-feedback-audios`
  - Purpose: Stores customer feedback audio files.

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6. Security

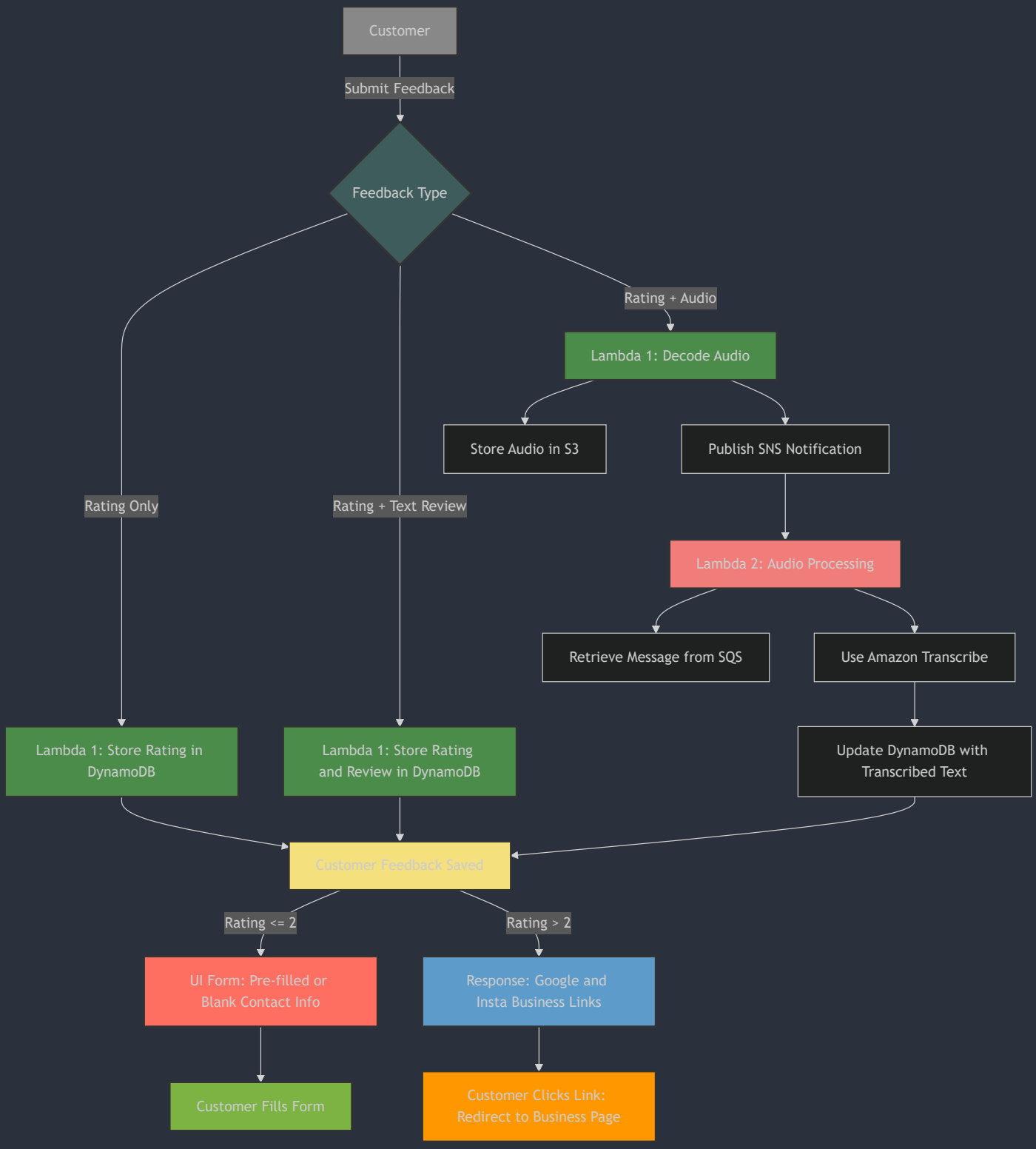
- **API Key Authentication**:
  - All API requests must include an API key in the `X-API-Key` header.

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7. Audience

This document is intended for **developers**, **testers**, and **integration engineers**. It provides:

- A comprehensive guide to the API endpoints.
- Instructions for debugging and testing.
- Integration tips for client applications.


Customer
Submit Feedback
Feedback Type
Rating Only
Rating + Text Review
Rating + Audio
Lambda 1: Decode Audio
Store Audio in S3
Publish SNS Notification
Lambda 2: Audio Processing
Retrieve Message from SQS
Use Amazon Transcribe
Update DynamoDB with Transcribed Text
Lambda 1: Store Rating in DynamoDB
Lambda 1: Store Rating and Review in DynamoDB
Customer Feedback Saved
Rating <= 2
Rating > 2
UI Form: Pre-filled or Blank Contact Info
Response: Google and Insta Business Links
Customer Fills Form
Customer Clicks Link: Redirect to Business Page